AI-Powered Agriculture Crop Advisory Assistant

Complete End-to-End System Architecture: Data -> RAG + ML -> Multi-Agent -> Guardrails -> Evaluation

1. DATA INGESTION & FEATURE ENGINEERING

Raw Dataset
data/raw/agriculture.csv

Preprocessing
Cleaning, Imputation, Norm

Case Construction
Structured JSON Documents

FAISS Index
Dense TF-IDF/SVD Vector

Keyword Index
Sparse BM25-style Tokens

ML Suitability
RandomForestClassifier

ML Yield Regressor
RandomForestRegressor + CI

RAG Evidence

ML Models

2. CLIENT & API LAYER

Web Dashboard
SPA (HTML/CSS/JS)

HTTP

FastAPI Gateway
REST Endpoints & Validation

Query

3. MULTI-STAGE GUARDRAILS

Input Guardrail
Injection, Scope & Range

4. 8-STAGE DETERMINISTIC MULTI-AGENT PIPELINE

Agent 1: Classifier
Intent & Parameter Extraction

Agent 2: Hybrid RAG
FAISS Dense + Sparse TF-IDF

Agent 3: Filter & Rerank
RRF Fusion + Agro-Scoring

Agent 4: Recommender
Grounded Crop Viability

Agent 5: Risk Agent
Drought, Salinity & Extremes

Agent 6: Yield Agent
ML Regressor + 80% CI Bounds

Agent 7: Impact Analysis
Feature Importance & Sensitivity

Agent 8: LLM-as-Judge
Groundedness Verification

Output Guardrails
Disclaimer & Safety Check

Deliver to User

5. ADVISORY RESPONSE, FEEDBACK & CONTINUOUS BENCHMARKING

Advisory Payload

Structured Advisory
Section 45 Agronomic Response

User Ratings

Feedback Store
data/feedback/feedback.json

Evaluation Benchmark
Precision@k, Recall@k, MRR, R^2

Evaluation Reports
reports/evaluation_report.md

[Legend] Light Blue: Data/RAG | Purple: ML Models | Green: Multi-Agents | Red: Guardrails | Amber: UI/API | Pink: Evaluation